

End-of-outbreak decision-making under superspreading and onset-date delays

The 2018 Équateur Province (DRC) Ebola outbreak

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Abstract

Deciding when an outbreak is over requires estimating the probability that further cases will occur if control measures are relaxed. This estimate depends on two often-implicit choices: how transmission heterogeneity is represented and whether symptom-onset dates are treated as if they were infection dates. We compared five renewal models of the 2018 Équateur Province Ebola outbreak, all calibrated to the same onset-to-onset serial interval, and calculated the risk of additional cases (RAC) after each day. Models placing overdispersion at the level of the day, transmission event, or individual produced materially different risks at the same nominal dispersion parameter and inferred different values when that parameter was estimated. Models that anchored transmission to the infector’s symptom onset and represented incubation explicitly crossed both decision thresholds 8–12 days earlier than their infection-anchored counterparts; they also inferred substantially smaller post-intervention reproduction numbers. Finally, distinguishing additional cases from additional transmission isolated the contribution of people already infected but not yet symptomatic—a component that infection-anchored models cannot represent. End-of-outbreak assessments therefore depend not only on the amount of transmission heterogeneity, but also on its mechanism and on the time scale to which transmission is anchored.

1 Introduction

An end-of-outbreak declaration is a decision under uncertainty: interventions are withdrawn when the probability of further cases is judged acceptably small. Renewal models provide a natural basis for that decision by linking observed incidence to past transmission and projecting forward under relaxed control. Two choices within these models are seldom examined together.

The mechanism of overdispersion. Transmission of Ebola virus disease, like that of many pathogens, is highly heterogeneous, and this is usually represented by a single dispersion parameter k . Models can share a mean–variance relationship at the population level while differing in *where* the excess variance sits — in a day’s aggregate incidence, in each transmission event, or in each infected individual. Those are different statements about the world, and we show they imply different end-of-outbreak risks. They also make k a different quantity in each case, so a literature estimate obtained under one mechanism is not portable to another. In particular, an individual-level estimate such as $k = 0.18$ does not retain its interpretation in a day-level incidence model.

Onset dates are not infection dates. Renewal models are routinely fitted with a symptom-onset time series substituted directly for an infection time series. This conflates two distinct events separated by an incubation period, which for Ebola virus disease has a mean of about 11.4 days. We quantify the consequences by comparing infection-anchored models with explicitly onset-anchored counterparts, in which transmission is referenced to the infector’s own onset and the incubation period is modelled.

We compare five models using the 2018 Équateur Province Ebola outbreak. All use the same onset-to-onset serial interval, allowing differences to be attributed to transmission mechanism and temporal anchoring rather than to different interval assumptions. We first compare three infection-anchored mechanisms with the dispersion fixed and estimated, then compare event- and individual-level models with onset-anchored counterparts. The primary estimand is the risk of additional cases; a supplementary analysis separates it from the risk of additional transmission. Sensitivity analyses needed to assess the robustness of these comparisons are specified separately.

2 Data and delay distributions

2.1 The 2018 Équateur outbreak

The observed series is a daily count of Ebola virus disease cases by *symptom-onset date* for the 2018 Équateur Province outbreak in the Democratic Republic of the Congo: 54 cases, with onsets on 31 distinct days between 5 April 2018 and 2 June 2018, and at most 6 onsets on any one day. An Ebola Response Team (ERT) was present from 8 May 2018 and withdrew on 24 July 2018. Indexing days from the first observed onset:

	Day index	Date
First onset (index case)	0	5 April 2018
ERT arrival	33	8 May 2018
Last observed onset	58	2 June 2018
ERT withdrawal (window end)	110	24 July 2018

The analysis window is days 0–110, i.e. 111 days, with 28 cases before the ERT arrived and 26 after. The series is padded with zero counts from the last observed onset to the end of the window. These case-free days provide the evidence that drives the estimated risk down. The main analyses take the series at face value; Section 6.5 relaxes that and treats the observed onsets as a reported fraction of the true ones.

Day 0 is treated as an initial condition in every model, so likelihoods run over days $1, \dots, 110$. Conditioning on a single imported index case asserts that nothing was transmitting or incubating beforehand. For the infection-anchored models this is mild — the renewal recursion simply starts from one case. For the onset-anchored models it is stronger, because it also asserts an *empty incubation pipeline* at day 0: anyone infected shortly before the index case’s onset would contribute onsets in the first weeks that the model can explain only as transmission from the index case, inflating R_{pre} . The onset-anchored models are therefore more sensitive to whether the first observed case is the true epidemiological index case.

2.2 Serial interval

We use a generic Ebola serial-interval estimate, represented by a discretised gamma distribution with mean 15.3 d and standard deviation 9.3 d, in all five models. An outbreak-specific estimate fitted to contact-traced pairs from this outbreak has mean 19.46 d and standard deviation 6.08 d.

We selected the generic estimate because it is compatible with the variance budget of Section 2.3. Splitting an onset-to-onset interval into an incubation period and a time-from-onset-to-transmission (TOST) delay requires the two variances to sum to the serial interval’s. The generic estimate has variance 86.49 d², which comfortably accommodates standard published Ebola incubation estimates. The outbreak-specific estimate has variance 36.97 d², *smaller* than the incubation variance alone (65.61 d²), and is therefore structurally inadmissible for the onset-anchored models under the assumed independent delay decomposition. Sensitivity to the outbreak-specific estimate can therefore be assessed only for the infection-anchored models.

2.3 Discretisation, indexing and the decomposition

Continuous delays are gamma distributed and discretised by the method of Cori et al. (2013): the mass at integer lag s is $\int_{s-1}^{s+1} (1 - |s - u|) g(u) du$, normalised. Three discrete delays are used, and they differ in the lag at which they start:

Distribution	Weights	First lag	Interpretation
Serial interval	w_s	1	onset of infector $\rightarrow$ onset of infectee
Incubation period	f_a^{inc}	1	infection $\rightarrow$ onset
Onset to transmission	f_s^{tost}	0	onset of infector $\rightarrow$ transmission

Write $F_r = \sum_{s=1}^r w_s$ with $F_0 = 0$ for the serial-interval distribution function, and let

$$\bar{f}_r^{\text{tost}} = \Pr(S > r) = 1 - \sum_{s=0}^r f_s^{\text{tost}}, \quad \bar{f}_r^{\text{inc}} = \Pr(A > r) = 1 - \sum_{a=1}^r f_a^{\text{inc}} \quad (1)$$

be the survival functions of the TOST delay S and the incubation period A . Note $\bar{f}_0^{\text{inc}} = 1$ but $\bar{f}_0^{\text{tost}} = 1 - f_0^{\text{tost}} \leq 1$: a case has all of its incubation ahead of it on the day it is infected, but has already had its lag-0 transmission opportunity on the day it becomes symptomatic.

The lag-0 rule is what makes the onset-anchored recursion causal. Because f^{inc} places no mass at lag 0, an infection on day t cannot produce an onset before day $t + 1$, so a day's onset cohort is fully determined before that day's own transmission is evaluated. The TOST distribution *may* place mass at lag 0, so a case can transmit on its own onset day. At most one of the two may be supported at lag 0: if both were, infection, onset and onward transmission could all fall within one day and the recursion would not be well ordered.

The variance budget. For an infector with onset on day u , transmission occurs on day $u + S$ and the infectee's onset on day $u + S + A$, so the onset-to-onset interval is $S + A$ and matching the target serial interval requires

$$\mathbb{E}[S] + \mathbb{E}[A] = 15.3, \quad \text{Var}[S] + \text{Var}[A] = 86.49. \quad (2)$$

The incubation period is fixed from a published Ebola estimate — mean 11.4 d, standard deviation 8.1 d (WHO Ebola Response Team, 2014) — and f^{tost} is the gamma carrying the residual mean and variance, here mean 3.9 d and standard deviation 4.57 d. A negative residual variance would make the incubation estimate inadmissible against the serial interval. Discretising and then convolving is not quite the same operation as convolving and then discretising, so we verified the decomposition numerically: the discrete convolution $f^{\text{tost}} * f^{\text{inc}}$ differs from the discretised target w by 0.00070 in total variation. All delays are truncated at lag 110, the length of the analysis window, so nothing is truncated within it (Fig. S1).

Remark. The convolution is faithful, but *how* the mean and variance are split between incubation and TOST is assumed rather than identified by the serial interval alone. The sensitivity of the onset-anchoring results to this decomposition must therefore be assessed explicitly.

3 Models

3.1 Notation

Let C_t be the observed count on day t . The three *infection-anchored* models set $I_t := C_t$, reading onset dates as infection dates — the practice under study. The two *onset-anchored* models set $D_t := C_t$ and treat onsets as onsets, with infections and onsets separated by the incubation period. The reproduction number R_t takes the value R_{pre} before the ERT arrives and R_{post} from its arrival onwards (Section 3.5), and $k > 0$ is a dispersion parameter, small k corresponding to high heterogeneity.

Remark. $\text{NB}(\text{mean} = \mu, \text{disp} = \kappa)$ denotes the negative binomial with mean μ , variance $\mu + \mu^2/\kappa$ and probability generating function $G(s) = (\kappa/(\kappa + \mu(1-s)))^\kappa$; in particular $G(0) = (1 + \mu/\kappa)^{-\kappa}$. $\text{Gamma}(\alpha, \beta)$ has shape α , rate β and mean α/β .

3.2 Infection-anchored models

The Poisson baseline and the three compared infection-anchored models share one generative skeleton. Cases beget cases through the serial interval: an individual infected on day u produces secondary infections that appear on day $u + \ell$ with probability w_ℓ , and the reproduction number in force on the day a secondary case appears sets their expected number. Write

$$\lambda_t = \sum_{s \geq 1} w_s I_{t-s} \quad (3)$$

for the force of infection driving day t . What distinguishes the models is the offspring law — where, if anywhere, the excess variance sits.

From transmission to observations. Infection and symptom onset are identified, so $I_t = C_t$. The transmission law for I_t is therefore also the likelihood for the observed daily counts; no incubation step intervenes.

Poisson baseline: Cori. An individual infected on day u generates, at each lag ℓ and independently of every other, a Poisson number of offspring with mean $R_{u+\ell}w_\ell$. Poisson superposition then gives the day- t total as

$$\text{Cori :} \quad I_t \mid \lambda_t \sim \text{Poisson}(R_t \lambda_t). \quad (4)$$

This is the standard renewal model of Cori et al. (2013) and the model underlying most published end-of-outbreak risk calculations. It has no dispersion parameter.

Transmission-event heterogeneity: SSE. The offspring law is negative binomial, with a dispersion that *does* scale with the force of infection:

$$\text{SSE :} \quad I_t \mid \lambda_t \sim \text{NB}(\text{mean} = R_t \lambda_t, \text{disp} = k \lambda_t). \quad (5)$$

That scaling is what makes k an *individual-level* offspring dispersion. Each infector independently contributes $\text{NB}(R_t w_\ell, k w_\ell)$ offspring at lag ℓ ; these all share the success probability $k/(k + R_t)$, which does not depend on ℓ , so by negative binomial closure they pool into exactly (5), and each individual's own offspring total is $\text{NB}(R, k)$. The decomposition is used again in Section 4.2.

Equivalently, (5) has the Poisson–Gamma representation

$$\tilde{\lambda}_t \mid \lambda_t \sim \text{Gamma}(k \lambda_t, k), \quad I_t \mid \tilde{\lambda}_t \sim \text{Poisson}(R_t \tilde{\lambda}_t), \quad (6)$$

in which the day's transmission opportunity is inflated or deflated by a fresh Gamma factor. The marginal form (5) is used directly for SSE; the mixture representation is needed to define its onset-anchored counterpart below.

Individual heterogeneity: SSI. Heterogeneity is attached to people rather than to occasions. Each infected individual carries a latent infectivity drawn independently from $\text{Gamma}(k, k)$, with mean 1 and variance $1/k$, so the aggregate infectivity of the day- t cohort is $\text{Gamma}(k I_t, k)$ and infections arise as a Poisson process driven by past infectivity:

$$\text{SSI :} \quad Y_t \mid I_t \sim \text{Gamma}(k I_t, k), \quad I_t \mid (Y_u)_{u < t} \sim \text{Poisson}(R_t \lambda_t^Y), \quad \lambda_t^Y = \sum_{s \geq 1} w_s Y_{t-s}. \quad (7)$$

Here $I_t = 0$ forces $Y_t = 0$. Marginally each individual's offspring total is $\text{NB}(R, k)$, the same as under SSE — the two models are calibrated to the same individual-level offspring distribution and differ only in mechanism. But the Y_t cannot be integrated out of the joint law and must be inferred with the other model parameters.

Day-level heterogeneity: DLO. Unlike the three families above, DLO has no onset-anchored counterpart in this study. The negative binomial is applied directly to the day's aggregate incidence, with the same dispersion on every day:

$$\text{DLO : } I_t \mid \lambda_t \sim \text{NB}(\text{mean} = R_t \lambda_t, \text{disp} = k). \quad (8)$$

Its excess variance is a property of the day and does not scale with the number of infectious people. Thus k describes day-to-day variation in aggregate incidence rather than inequality in individual transmission, and the same numerical value has a different meaning here than in SSE or SSI.

Remark (The Poisson limit). As $k \rightarrow \infty$ the negative binomial in (5) converges to the Poisson and the Gamma noise in (7) degenerates to $Y_t = I_t$, so SSE and SSI both collapse onto Cori. DLO loses its excess variance in the same limit, but without the parameter ever having had an individual-level reading to lose.

3.3 Onset-anchored models

Transmission is now referenced to the infector's own symptom onset rather than to their unobserved infection time, which is appropriate for a disease with negligible pre-symptomatic transmission. Two things change. The serial interval is replaced by the TOST distribution, so the force of infection is a weighted sum of past *onset* cohorts; and infection and onset become distinct events, separated by the incubation period. Each of the three models below is the exact onset-anchored counterpart of one of (4), (5) and (7), and is presented in the same order. DLO is omitted because no onset-anchored day-level model is compared.

From transmission to observations. Let J_t denote the number of infections on day t and $D_t = C_t$ the number of observed onsets. In all three models, day- u infections are carried forward to their onset days by independent incubation draws,

$$(N_{u,a})_{a \geq 1} \mid J_u \sim \text{Multinomial}(J_u, (f_a^{\text{inc}})), \quad D_t = \sum_{a \geq 1} N_{t-a,a}, \quad (9)$$

and only the onsets are observed. The infection counts J_t and allocations $N_{u,a}$ can be integrated out exactly.

Proposition 1 (Onsets alone suffice). *Suppose that, conditional on a non-negative day- t rate E_t that is a function of $D_{0:t}$ and of any latent variables attached to days $u \leq t$, the infections satisfy $J_t \mid E_t \sim \text{Poisson}(E_t)$, and that they are allocated to onset days by (9). Then that model and*

$$D_t \mid (E_u)_{u < t} \sim \text{Poisson}\left(\sum_{a \geq 1} f_a^{\text{inc}} E_{t-a}\right) \quad (10)$$

define the same joint law for the observed onset process (D_t) .

The proof is in Appendix A. It rests on Poisson thinning, which turns the multinomial split (9) into independent Poisson parts, and on Poisson superposition, which recombines them into (10); the same thinning argument supplies the incubation pipeline of Section 4.3. The hypothesis covers all three models: two have Poisson offspring directly, and SSE-SO's negative binomial is Poisson conditional on its Gamma factor, as in (6). We therefore give both the infection process and the equivalent likelihood for observed onsets.

Poisson baseline: Cori-SO. A case with onset on day u infects others s days later with probability f_s^{tost} , in $\text{Poisson}(R_{u+s}f_s^{\text{tost}})$ numbers and independently across lags and individuals. Summing over infectors,

$$\text{Cori-SO : } J_t \mid D_{0:t} \sim \text{Poisson}(R_t \lambda_t), \quad \lambda_t = \sum_{s \geq 0} f_s^{\text{tost}} D_{t-s}, \quad (11)$$

together with (9). Proposition 1 applies with $E_t = R_t \lambda_t$ and reduces this to

$$D_t \mid D_{0:t-1} \sim \text{Poisson} \left(\sum_{a \geq 1} f_a^{\text{inc}} R_{t-a} \sum_{s \geq 0} f_s^{\text{tost}} D_{t-a-s} \right), \quad (12)$$

a deterministic recursion in the observed onsets.

Transmission-event heterogeneity: SSE-SO. The offspring law is negative binomial with a dispersion scaling with the force of infection, as in (5):

$$\text{SSE-SO : } J_t \mid D_{0:t} \sim \text{NB}(\text{mean} = R_t \lambda_t, \text{disp} = k \lambda_t), \quad \lambda_t = \sum_{s \geq 0} f_s^{\text{tost}} D_{t-s}, \quad (13)$$

together with (9). Writing the negative binomial as a Poisson–Gamma mixture puts a fresh Gamma factor on each day’s transmission opportunity, and Proposition 1 then applies with $E_t = R_t \tilde{\lambda}_t$:

$$\tilde{\lambda}_t \mid \lambda_t \sim \text{Gamma}(k \lambda_t, k), \quad D_t \mid (\tilde{\lambda}_u)_{u < t} \sim \text{Poisson} \left(\sum_{a \geq 1} f_a^{\text{inc}} R_{t-a} \tilde{\lambda}_{t-a} \right). \quad (14)$$

Here, unlike in SSE, the Gamma factor cannot be integrated out again: each D_t pools contributions from many days, each carrying its own independent $\tilde{\lambda}$, so the onsets are *not* marginally negative binomial. The day-specific Gamma effects must therefore be retained in the likelihood.

Individual heterogeneity: SSI-SO. Each case carries a latent infectivity drawn from $\text{Gamma}(k, k)$, so the day- t onset cohort has aggregate infectivity $\text{Gamma}(k D_t, k)$ and drives a Poisson infection process, as in (7):

$$\begin{aligned} \text{SSI-SO : } Y_t \mid D_t &\sim \text{Gamma}(k D_t, k), & J_t \mid (Y_u)_{u \leq t} &\sim \text{Poisson}(R_t \lambda_t^Y), \\ \lambda_t^Y &= \sum_{s \geq 0} f_s^{\text{tost}} Y_{t-s}, \end{aligned} \quad (15)$$

together with (9). Proposition 1 applies with $E_t = R_t \lambda_t^Y$ and gives

$$Y_t \mid D_t \sim \text{Gamma}(k D_t, k), \quad D_t \mid (Y_u)_{u < t} \sim \text{Poisson} \left(\sum_{a \geq 1} f_a^{\text{inc}} R_{t-a} \sum_{s \geq 0} f_s^{\text{tost}} Y_{t-a-s} \right). \quad (16)$$

The latents are the same cohort infectivities as under SSI and, as there, cannot be integrated out.

Remark (The Poisson limit). As in Remark 3.2, $k \rightarrow \infty$ collapses SSE-SO and SSI-SO onto Cori-SO. Neither Cori nor Cori-SO is one of the five compared models. They provide the Poisson limits against which the overdispersed models are interpreted; Cori also links the analysis to a published Équateur risk curve.

3.4 The five compared models

Model	Anchoring	Excess variance acts at the level of	Latents
DLO	infections [†]	the day — NB on aggregate incidence	none
SSE	infections [†]	the transmission event	none
SSI	infections [†]	the individual	Y_t
SSE-SO	symptom onsets	the transmission event	$\tilde{\lambda}_t$
SSI-SO	symptom onsets	the individual	Y_t

[†]Observed onsets are treated as infection dates.

3.5 When does R switch?

R switches from R_{pre} to R_{post} on day 33 (8 May 2018) **in each model’s own time index**. In the onset-anchored models this is unambiguous: R_t multiplies the TOST-weighted force of infection producing *infections* on day t , so the intervention acts on transmission from that date and the resulting onsets appear an incubation period later. In the infection-anchored models R_t multiplies the force of infection producing *cases appearing* on day t , so the switch attributes the intervention’s effect to onsets from that date.

The displacement between the two conventions is the infectee’s **incubation period**, not the serial interval. A case with onset on day t was infected on day $t - A$ with $A \sim f^{\text{inc}}$; the infection-anchored model dates the intervention’s effect on that transmission to t , the onset-anchored model to $t - A$. The relevant delay is therefore $\mathbb{E}[A] = 11.4$ d. (The serial interval, 15.3 d, is the infector-onset to infectee-onset gap and does not separate the two conventions.)

This asymmetry is a consequence of onset-as-infection conflation: it is one route by which infection-anchored models mis-date an intervention’s effect. Section 6.3 quantifies its consequences for R_{post} and the risk curve.

4 Risks of additional cases, additional transmission and sustained transmission

4.1 Definition

The estimand is a real-time one, and we state it as a procedure.

Fit $(R_{\text{pre}}, R_{\text{post}}, k)$ and all latent states to the observed record through day t alone. Then discard the realised trajectory after day t , reset R to $R^* = R_{\text{pre}}$ from day $t + 1$ onwards, and simulate a counterfactual future. $\text{RAC}(t)$ is the posterior probability that this replicated future contains at least one further case.

So $\text{Pr}(\cdot \mid \text{data up to day } t)$ is the correct reading, and every conditioning day carries its own fit: the curve over days 1–110 is 110 fits per model. Day 0 is an initial condition in every model and contributes no likelihood term, so the curve begins at day 1. Section 4.5 explains why the parameters cannot be estimated once and reused.

One consequence is visible at the left-hand end of every risk curve below. In the opening days the record is a single case and a short run of silence, the reproduction number is prior-dominated, and the risk of at least one further case is therefore well below one; it climbs towards one as cases accumulate, and only then begins the descent the analysis is about. A retrospective estimand cannot show this, because parameters informed by the whole outbreak place the early risk at one by construction. The day-to-day roughness over those first weeks is equally a feature of the estimand rather than of the sampler: each day is a separate posterior conditioned on a separate record.

Under the onset-anchored models a second quantity becomes distinct:

- $\text{RAC}(t)$, the probability of at least one further *case*, i.e. a further entry in the onset series after day t . It counts individuals who are already infected on day t and become symptomatic afterwards, even if no new transmission occurs, so it does not approach zero until the incubation pipeline has drained. This is the directly observable quantity and the headline throughout.
- $\text{RAT}(t)$, the probability of at least one further *transmission event* after day t . It isolates the effect of relaxing interventions but is not directly observable.
- $\text{RST}(t)$, the probability that the counterfactual transmission process never becomes extinct. “Sustained” means eventual non-extinction, not reaching an arbitrary finite case threshold.

Under the three infection-anchored models an infection and the case it produces fall on the same day by construction, so the two coincide *by assumption* — which is precisely the conflation under study. RST is defined for the four models with individual offspring families and for their Poisson limits. It is not defined for DLO, whose day-level negative-binomial variation does not identify an individual’s descendants.

$\text{RAC}(t)$ is one number per day, not a distribution: it is a posterior probability, so the average over draws happens inside it. In each case below we obtain a conditional zero-probability given one posterior draw, and report

$$\text{RAC}(t) = 1 - \frac{1}{S} \sum_{m=1}^S \exp(\log \Pr(\text{no further case} \mid \theta^{(m)})), \quad (17)$$

over S posterior draws, and likewise for RAT and RST . Because this is an average it carries Monte-Carlo error, which is estimated from the between-chain spread and reported beside every curve.

4.2 Infection-anchored models

Fix one posterior draw. In the infection-anchored models, the observed incidence and, for SSI, the retained infectivities completely specify the reset state: there is no separate incubation pipeline.

Retained transmission weight. Define

$$\Lambda(t) = \sum_{u \leq t} I_u (1 - F_{t-u}) \quad (\text{Cori, SSE, DLO}), \quad \Lambda_Y(t) = \sum_{u \leq t} Y_u (1 - F_{t-u}) \quad (\text{SSI}), \quad (18)$$

where $(1 - F_{t-u})$ is the probability that a case first appearing on day u still has transmission to come after day t ; with $F_0 = 0$, the $u = t$ term enters with full weight.

Extinction under the reset. Let q be the smallest extinction fixed point of the offspring distribution of one case after the reset:

$$q = \left(1 + \frac{R^*}{k}(1 - q)\right)^{-k} \quad (\text{negative binomial}), \quad q = \exp\{-R^*(1 - q)\} \quad (\text{Poisson}). \quad (19)$$

At and below criticality, $R^* \leq 1$, the smallest fixed point is $q = 1$ exactly.

Future-event exponent. Let $H_{\text{inf}}(t)$ be minus the log-probability that the retained state produces no future event:

$$H_{\text{inf}}(t) = \begin{cases} R^* \Lambda(t) & (\text{Cori}), \\ k \Lambda(t) \log(1 + R^*/k) & (\text{SSE}), \\ R^* \Lambda_Y(t) & (\text{SSI}), \\ k \sum_{j>t} \log(1 + R^* \mu_j/k) & (\text{DLO}), \end{cases} \quad \mu_j = \sum_{s \geq 1} w_s I_{j-s}, \quad (20)$$

where the future counts in μ_j are set to zero, $I_u := 0$ for $u > t$.

Proposition 2 (Conditional zero-probabilities, infection-anchored models). *Conditional on one draw of the parameters and, for SSI, the retained latent infectivities,*

$$\log \Pr(\text{no further case after } t) = -H_{\text{inf}}(t), \quad (21)$$

$$\log \Pr(\text{no further transmission after } t) = -H_{\text{inf}}(t). \quad (22)$$

$$\log \Pr(\text{no sustained transmission after } t) = \begin{cases} \Lambda(t) \log q & (\text{SSE}), \\ -R^*(1-q)\Lambda_Y(t) & (\text{SSI}), \\ -R^*(1-q)\Lambda(t) & (\text{Cori}). \end{cases} \quad (23)$$

Proof. Retained cohorts. For Cori, each observed case contributes $\text{Poisson}(R^*w_\ell)$ offspring at lag ℓ independently; the contributions with $\ell > t - u$ are untouched by the conditioning and pool by Poisson superposition into $\text{Poisson}(R^*\Lambda(t))$. For SSE, the same contributions are $\text{NB}(R^*w_\ell, kw_\ell)$ and pool by negative binomial closure into $\text{NB}(R^*\Lambda(t), k\Lambda(t))$, whose zero-probability is as stated. For SSI, given the latents they are again Poisson, with rates $R^*w_\ell Y_u$, pooling into $\text{Poisson}(R^*\Lambda_Y(t))$. For DLO, the future counts are negative binomial day by day with a dispersion that does not depend on the force of infection, so the zero-probabilities multiply over days. Taking minus the logarithm gives the four rows of (20).

Cases versus transmission. Because infection and symptom onset are identified in these models, a future transmission event and the future case it creates are represented as occurring together.

Eventual extinction. Condition on the number N of future infections seeded by the retained state. Each begins an independent family that becomes extinct with probability q , so the required probability is $\mathbb{E}[q^N]$. Under SSE, N is $\text{NB}(R^*\Lambda(t), k\Lambda(t))$ and its pgf at q is $(1 + R^*(1-q)/k)^{-k\Lambda(t)} = q^{\Lambda(t)}$ by (19). Under SSI it is conditionally $\text{Poisson}(R^*\Lambda_Y(t))$, and under Cori it is $\text{Poisson}(R^*\Lambda(t))$; their pgfs give the remaining rows of (23). DLO has no row because it defines aggregate incidence one day at a time rather than individual offspring families. $\square$

Thus $\text{RAC}(t) = \text{RAT}(t) = 1 - e^{-H_{\text{inf}}(t)}$, while RST follows (23). The Cori row is included as the Poisson reference, not as a compared model. It is the $k \rightarrow \infty$ limit of the SSE and SSI rows, with $\Lambda_Y(t) \rightarrow \Lambda(t)$. The DLO sum is finite because $\mu_j = 0$ once j exceeds t plus the support of w .

4.3 Onset-anchored models

Fix one posterior draw. Here the observed onsets do not completely specify the reset state: in addition to retained onset cohorts, it contains a latent pipeline of individuals already infected but not yet symptomatic. Write R_u for the historical reproduction number on day u .

Retained transmission weight. The driving series still able to transmit after t ,

$$W(t) = \sum_{u \leq t} \mathcal{D}_u \bar{f}_{t-u}^{\text{tost}}, \quad \mathcal{D}_u = \begin{cases} D_u & (\text{Cori-SO}), \\ D_u & (\text{SSE-SO}), \\ Y_u & (\text{SSI-SO}), \end{cases} \quad (24)$$

the onset-anchored analogue of $\Lambda(t)$, with the TOST survival function of (1) in place of $1 - F$.

Incubation pipeline mean. The expected number infected on or before day t whose onset falls after it,

$$M(t) = \sum_{u \leq t} E_u \bar{f}_{t-u}^{\text{inc}}, \quad (25)$$

with E_u the model's conditional mean number of infections on day u — $R_u \lambda_u$ for Cori-SO, $R_u \tilde{\lambda}_u$ for SSE-SO and $R_u \lambda_u^Y$ for SSI-SO, as in (12)–(16) — evaluated at the *historical* R_u : the pipeline was filled before the reset.

Future-event exponent. Let c be minus the log-probability that one future case, transmitting under R^* over its whole TOST support, has no offspring:

$$c = \begin{cases} R^* & (\text{Cori-SO}), \\ k \log(1 + R^*/k) & (\text{SSE-SO}), \\ k \log(1 + R^*/k) & (\text{SSI-SO}). \end{cases} \quad (26)$$

The exponent for no transmission from onset cohorts already observed by day t is

$$H_{\text{on}}(t) = \begin{cases} R^* W(t) & (\text{Cori-SO}), \\ c W(t) & (\text{SSE-SO}), \\ R^* W(t) & (\text{SSI-SO}). \end{cases} \quad (27)$$

For SSI-SO, $H_{\text{on}}(t)$ conditions on the inferred infectivities of onset cohorts already observed, whereas c averages over the as-yet-unobserved infectivity of a future pipeline member.

Proposition 3 (Conditional zero-probabilities, onset-anchored models). *Conditional on one draw of the parameters and the retained latent state,*

$$\log \Pr(\text{no further case after } t) = -H_{\text{on}}(t) - M(t), \quad (28)$$

$$\log \Pr(\text{no further transmission after } t) = -H_{\text{on}}(t) - M(t)(1 - e^{-c}). \quad (29)$$

$$\log \Pr(\text{no sustained transmission after } t) = \begin{cases} W(t) \log q - M(t)(1 - q) & (\text{SSE-SO}), \\ -R^*(1 - q)W(t) - M(t)(1 - q) & (\text{SSI-SO, Cori-SO}). \end{cases} \quad (30)$$

Proof. Two independent sources of future events must be accounted for: the onset cohorts retained through day t , which have transmission opportunity left, and the pipeline of individuals already infected but not yet symptomatic.

Retained cohorts. Under Cori-SO and SSI-SO the infections generated by cohort u at TOST lag s are independent $\text{Poisson}(R^* f_s^{\text{tost}} \mathcal{D}_u)$; those with $s > t - u$ are untouched by the conditioning, and summing over s and $u \leq t$ gives $\text{Poisson}(R^* W(t))$, with zero-probability $e^{-R^* W(t)}$. Under SSE-SO the same contributions are $\text{NB}(R^* f_s^{\text{tost}} \mathcal{D}_u, k f_s^{\text{tost}} \mathcal{D}_u)$ with common success probability $k/(k + R^*)$, so they pool into $\text{NB}(R^* W(t), kW(t))$, whose zero-probability is $(1 + R^*/k)^{-kW(t)} = e^{-cW(t)}$. Either way the probability is $e^{-H_{\text{on}}(t)}$.

Pipeline. Poisson splitting of the day- u infections by the incubation delay (the argument of Appendix A) makes the number infected on or before t but symptomatic after t Poisson with mean $M(t)$, independently of the retained cohorts' remaining transmission.

Cases versus transmission. A future case requires either a pipeline member to become symptomatic or a retained cohort to generate a new infection — and every new infection produces an onset strictly later, since f^{inc} has no mass at lag 0. So no further case has probability $e^{-M(t)} e^{-H_{\text{on}}(t)}$, giving (28).

A future transmission event permits pipeline members to become symptomatic provided each transmits zero times, which each does independently with probability e^{-c} by (26). If the pipeline size is $N \sim \text{Poisson}(M(t))$ then $\mathbb{E}[(e^{-c})^N] = \exp(-M(t)(1 - e^{-c}))$ by the Poisson generating function, and multiplying by $e^{-H_{\text{on}}(t)}$ gives (29).

For no sustained transmission, every future infection must instead start a family that eventually becomes extinct. The retained SSE-SO cohorts seed a negative-binomial number of infections whose pgf at q is $q^{W(t)}$ by (19); the retained SSI-SO and Cori-SO cohorts seed a conditionally Poisson number, giving $\exp\{-R^*(1 - q)W(t)\}$. The pipeline contains $N \sim \text{Poisson}(M(t))$ cases and contributes $\mathbb{E}[q^N] = \exp\{-M(t)(1 - q)\}$ in every model. Multiplication gives (30). $\square$

Comparison of the two anchorings. Both $H_{\text{inf}}(t)$ and $H_{\text{on}}(t)$ describe transmission still owed by cohorts observed through day t . The onset-anchored calculation adds $M(t)$, the infections already generated under the historical R_u whose onsets lie after t . A further *case* requires that pipeline to be empty; no further *transmission* requires only that its members have no offspring. Setting $M(t) = 0$ reduces the onset-anchored expressions to the same retained-cohort structure as their infection-anchored counterparts, with the TOST survival replacing the serial-interval survival.

Remark (Immediate consequences). Five special cases follow immediately and are used as verification targets (Section 5.6).

- (i) Since $1 - e^{-c} \leq 1$, (29) is never below (28), so $\text{RAC}(t) \geq \text{RAT}(t)$ always, with equality exactly when $M(t) = 0$. The gap is the pipeline, and it is the term the infection-anchored models set to zero by assumption.
- (ii) A sustained family requires a transmission event, so $\text{RST}(t) \leq \text{RAT}(t) \leq \text{RAC}(t)$. Equivalently, the one-case extinction probability q is at least its zero-offspring probability e^{-c} , and the same ordering follows term by term in (29) and (30).
- (iii) Setting $R^* = 0$ gives $c = 0$ and $H_{\text{on}}(t) = 0$, so $\text{RAT}(t) = 0$ exactly and $\text{RAC}(t) = 1 - e^{-M(t)}$: with no future transmission permitted, the only remaining risk is that the retained pipeline is non-empty.
- (iv) As $k \rightarrow \infty$, $c = k \log(1 + R^*/k) \rightarrow R^*$ and SSE-SO's $H_{\text{on}}(t)$ tends to $R^*W(t)$, so both overdispersed models collapse onto Cori-SO.
- (v) If $R^* \leq 1$, then $q = 1$ and $\text{RST}(t) = 0$ exactly, whatever reset state remains.

Remark (Why time delays do not change eventual extinction). The serial-interval and TOST distributions are normalised: they allocate a case's offspring across future times without changing its total offspring pgf. They therefore change $\Lambda(t)$, $W(t)$ and $M(t)$, and hence the timing of the decline in RST , but not the fixed point q under a constant reset reproduction number.

Remark (Why historical and reset reproduction numbers both appear). $H_{\text{on}}(t)$ uses the reset R^* , because it describes transmission that has yet to happen; $M(t)$ uses the historical R_u , because the pipeline it counts was generated before the reset. Under the infection-anchored models there is no pipeline, so the whole retained state is projected at R^* . This is the formal counterpart of the timing argument of Section 6.3.

4.4 What the three mechanisms do differently

Proposition 2 makes the structural comparison explicit. SSE and SSI depend on the observed history *only through the scalar* $\Lambda(t)$ or $\Lambda_Y(t)$: the total remaining offspring pools into a single negative binomial or, given the latents, a single Poisson. DLO does not. Its risk depends on the whole temporal *profile* $(\mu_j)_{j>t}$ of the future force of infection, because a fresh dispersion that does not scale with that force is applied on each future day. Writing $\varphi(\mu) = -k \log(1 + R^*\mu/k)$, which is convex with $\varphi(0) = 0$ and hence superadditive,

$$-R^*\Lambda(t) \leq \log \Pr(\text{no further case})_{\text{DLO}} = \sum_{j>t} \varphi(\mu_j) \leq \varphi(\Lambda(t)). \quad (31)$$

Spreading the same total $\Lambda(t)$ across days *dilutes* the overdispersion, so a value of k calibrated as an individual-level offspring dispersion does far less work in DLO than in SSE or SSI. The lower bound is the negative of the Cori exponent in (20), so DLO's risk is squeezed towards Cori's precisely when the future force of infection is thin and spread out.

Remark. The bounds (31) hold always; the *direction* of the comparison with SSE does not. It is a property of a thin, spread-out profile rather than a theorem, and concentrating $\Lambda(t)$ into a day or two reverses it.

4.5 Why every conditioning day carries its own fit

Real-time conditioning applies to the parameters as well as to the latent states, and the two cannot be separated. Under SSI, SSE-SO and SSI-SO the reproduction numbers and the latent infectivities are correlated in the posterior: a history is explained either by a higher R or by more infectious cohorts, and the data through day t constrain that trade-off jointly. Retaining a latent state conditioned on the first t days beside parameters conditioned on the whole record is therefore not the day- t posterior of either, and refitting is what the estimand requires rather than a refinement of it.

The alternative is nonetheless informative, and we report it as a comparison rather than adopting it: one fit to the complete record, with the latent state re-derived at each day by a bootstrap particle filter run at every parameter draw. DLO and SSE have no latent state, so for them that comparison isolates the effect of conditioning the parameters alone, which makes them a useful control when reading it.

Only the conditioning changes. Because $\Lambda(t)$, $W(t)$ and $M(t)$ sum over days $u \leq t$, truncating the record to day t leaves the risk formulas of Propositions 2 and 3 untouched.

4.6 Integrating out the remaining latent states

Some latent states are integrated out of the likelihood analytically (Section 5.3) but still enter the reset state — in particular infections after the final observed onset, which may remain in the incubation pipeline. They need not be recovered individually. In every model the conditional log zero-probability is *affine* in the latent trajectory,

$$\log \Pr(\text{no further event after } t \mid \theta, Y) = -\left(\alpha(\theta) + \sum_u \beta_u(\theta) Y_u\right), \quad \beta_u \geq 0, \quad (32)$$

and the removed states are conditionally independent $\text{Gamma}(A_u, B_u)$ given θ , with $A_u = k \text{scale}_u$ and $B_u = k + c_u$. The Gamma moment generating function therefore integrates them out of the risk exactly,

$$\mathbb{E}[e^{-\beta_u Y_u} \mid \theta] = (1 + \beta_u / B_u)^{-A_u}, \quad (33)$$

so no Monte-Carlo reconstruction step is involved and the risk carries no sampling error beyond that of the posterior draws themselves. This is a third exact marginalisation, composing with the two already performed: the likelihood integrates these states out of the observation density, and (33) integrates the same states out of the risk. For RST, the retained-infectivity coefficient is $R^*(1 - q)$ and every historical incubation-pipeline coefficient is multiplied by $(1 - q)$; the existing RAC and RAT coefficients are unchanged.

5 Inference

5.1 Priors

We use proper priors, shared across models within each analysis, so that marginal likelihoods and posterior model probabilities are well defined.

- R_{pre} and R_{post} : independent log-normals with median 1.0 and 95% interval (0.2, 5.0).
- k , where estimated: an informative log-normal with median 0.18 and 95% interval (0.09, 0.36), identical in every model.
- Prior model probabilities: uniform over the models in each analysis, i.e. 1/3 in Analyses 1–2 and 1/4 in Analyses 3–4.

Remark (Identical priors are necessary but not sufficient for a fair comparison). Identical priors do not imply identical interpretations. In SSE and SSI, k is an individual-level offspring dispersion; in DLO, it controls day-level incidence variation (Section 4.4). Analysis 1 applies an individual-level

literature estimate to all three models to test its portability. Its posterior model probabilities therefore compare the full model specifications, including that imposed value, and should not be read as evidence about mechanism alone.

5.2 Posterior sampling

All fits use PyMC, with 4 chains of 2000 draws after 2000 tuning iterations in every analysis. Fits with fixed k , and estimated- k fits without an inverse-CDF latent block, use Hamiltonian Monte Carlo throughout (the No-U-Turn sampler, NUTS). In estimated- k fits of SSI, SSE-SO and SSI-SO, the Gamma quantile is differentiable with respect to its probability coordinate but not its shape. PyMC therefore uses a compound kernel: univariate Slice sampling for k , and NUTS for R_{pre} , R_{post} and the latent Uniform coordinates. This is the same compound sampler evaluated by the parameterisation benchmark. The NUTS target acceptance probability is 0.90 in Analyses 1–3 and 0.95 in Analysis 4. Each fit jointly estimates R_{pre} , R_{post} , any latent states, and — in Analyses 2 and 4 — k . Because the estimand conditions on the record through the day in question (Section 4.5), each model is fitted once per conditioning day, and the diagnostics below cover all of those fits. The evidence and dispersion comparisons are properties of a model given the whole record, and use the fit to the complete series, which also serves as the final conditioning day.

Convergence. The four analyses rest on 1852 fits in total. Across every one of them, and over every sampled variable including the latent blocks, the worst $\hat{R}$ is 1.017, the smallest bulk effective sample size is 248, and the total number of divergent NUTS transitions is 174. Per analysis the minimum bulk effective sample sizes are 3666, 3180, 1020 and 1407. The analysis workflow refuses to produce a risk curve if any of the fits behind it exceeds $\hat{R} = 1.01$ or shows divergent transitions in more than 1% of draws, so these are enforced criteria rather than a report of what happened to occur.

5.3 Latent-state sampling

The Gamma latent variables, particularly in SSE-SO, include very small shape parameters. We combine exact marginalisation with an inverse-CDF parameterisation to obtain stable posterior sampling without changing the model.

- **Exact marginalisation.** A latent that reaches no observation day carrying a case enters the likelihood only through an exponential factor linear in that latent. Its Gamma prior is conjugate to this factor, giving $\int \text{Gamma}(y; k \text{ scale}, k) e^{-cy} dy = (1 + c/k)^{-k \text{ scale}}$ and a conditional posterior $\text{Gamma}(k \text{ scale}, k + c)$. This removes 52 of SSE-SO’s 110 latents, leaving 58 sampled, and removes 1 of SSI’s and SSI-SO’s 31 on the complete record, and more on the shorter records of earlier conditioning days. Section 4.6 shows how they re-enter the risk calculation exactly.
- **Inverse-CDF reparameterisation.** The surviving latents are sampled as $\text{Uniform}(0, 1)$ variates transformed through the corresponding Gamma quantile function.

5.4 Model evidence

We compute the marginal likelihood $p(D_{1:110} \mid \text{model})$ by integrating over both parameters and latent states. Bridge sampling is the primary estimator; prior Monte Carlo and importance sampling provide independent numerical checks, and deterministic quadrature is used where the parameter space is sufficiently small. Posterior model probabilities follow from these marginal likelihoods and the uniform model prior of Section 5.1.

5.5 Comparing dispersion posteriors

Where k is estimated, we compare every pair of posterior distributions using three complementary quantities: the *median ratio*, which measures location; the *overlap* $\int \min(p_a, p_b)$, a scale-free symmetric measure; and $\Pr(k_a > k_b)$ for independent draws. All are computed on the log scale, as the density panels are drawn; the overlap is invariant to that choice because both densities acquire the same Jacobian.

5.6 Model and numerical validation

We validated the models against analytic limits and independent simulation. A deterministic one-day incubation period recovers the infection-anchored models after shifting the time index; as $k \rightarrow \infty$, SSE and SSI converge to Cori, while SSE-SO and SSI-SO converge to Cori-SO. Simulations from the explicit infection-and-incubation process agree with the onset-only representation of Proposition 1. Explicit reset simulations also agree with the risk formulas in Propositions 2 and 3, including the $R^* = 0$ limit in which $\text{RAT} = 0$ and RAC is the probability that the incubation pipeline is non-empty. The extinction roots are checked against their offspring generating functions, including the negative-binomial-to-Poisson limit, and the RST formulas are compared with independent Galton–Watson continuation simulation for all four branching models and both Poisson limits.

Particle filtering provides an independent check of the latent-state calculations, and particle marginal Metropolis–Hastings reproduces the parameter posteriors on synthetic data. Comparisons against particle methods match the conditioning on both sides: at fixed parameters the closed forms are held against a particle *smoother*, while the filtering states target the comparison estimator of Section 4.5. The affine representation (32) is verified against the risk formulas by the identity $\beta_u = \log \Pr(\cdot \mid Y = 0) - \log \Pr(\cdot \mid Y = e_u)$, which linearity makes exact, and the resulting marginalisation (33) is checked against averaging explicit draws from the same conditional distribution. Bridge-sampling estimates agree with deterministic quadrature for the low-dimensional models and with alternative estimators for the latent models. Finally, in the Poisson limit the analysis recovers the published reproduction-number posterior and Équateur risk curve of Thompson et al. (2024), after accounting for the one-day difference in time-indexing convention.

6 Results

Four analyses are reported. Analyses 1 and 2 compare the three infection-anchored models, with k held at the literature value 0.18 and then estimated; Analyses 3 and 4 add the onset-anchored models to SSE and SSI, again fixed and then estimated. Throughout, “the crossing” means the first day on which a curve falls below a threshold *and stays there*. Values quoted at day 90 (4 July 2018) are read from the fitted curves partway down their descent, where they are furthest apart.

6.1 Mechanism, at a transplanted dispersion parameter

- **The same numerical k produces little overdispersion in DLO.** At day 90 the risk is 0.412 under DLO against 0.093 under SSE and 0.123 under SSI — a factor of more than four between models handed an identical dispersion parameter. This is the dilution bound (31) in practice: the future force of infection after the last case is thin and spread over the whole serial interval, which is exactly the regime in which a day-level k does almost nothing and DLO is pressed against its Poisson lower bound.
- **The consequence for a declaration is weeks, not days.** DLO’s risk first settles below 0.05 on day 107 (21 July 2018), against day 95 for SSE and day 97 for SSI. At the 0.01 threshold DLO **never** settles within the analysis window, while SSE does so on day 106 and

Équateur 2018: naive (onset-as-infection) models, k fixed at 0.18

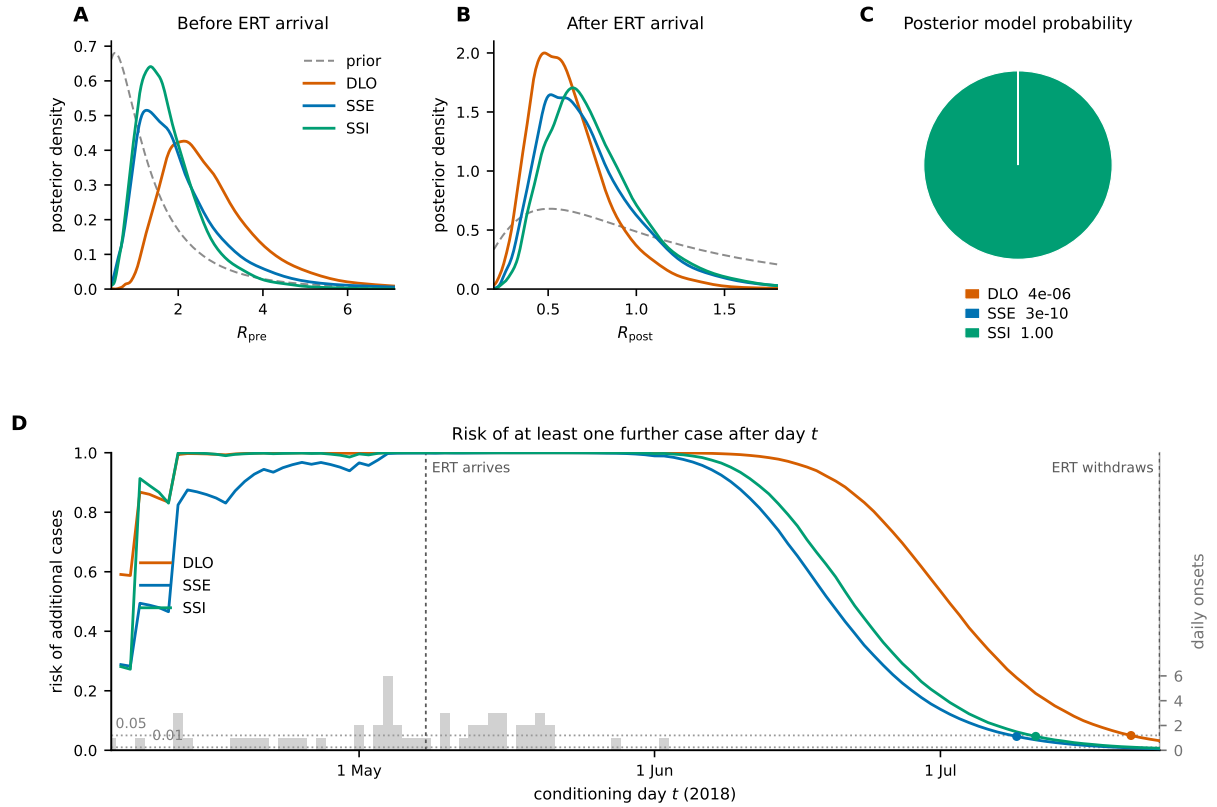


Figure 1: The mechanism of overdispersion changes the end-of-outbreak risk, at a dispersion parameter whose interpretation differs across models. DLO, SSE and SSI fitted to the Équateur onset series with k fixed at 0.18. **A, B:** posterior densities for R_{pre} and R_{post} , over the shared prior. **C:** posterior model probabilities under a uniform prior over the 3 models. Because the fixed value of k has a different interpretation in DLO, this panel compares the complete model specifications rather than mechanism alone. **D:** risk of additional cases against conditioning day, with daily onsets behind it, the ERT arrival and withdrawal marked, and the 0.05 and 0.01 decision thresholds drawn. The marker on each curve is its 0.05 crossing.

SSI on day 108. On the day the ERT actually withdrew, DLO still places 0.032 on a further case, against 0.005 and 0.007.

- **SSI lies on the same side of SSE as DLO does.** After the final case SSI’s risk is *above* SSE’s (0.123 against 0.093 at day 90), which reads backwards until one asks what each model’s heterogeneity is attached to. SSI’s is a property of *people who have been observed*: 54 known individuals have now gone weeks without infecting anyone, which is direct evidence they were not the superspreading kind, and $\Lambda_Y(t)$ is revised down accordingly. SSE’s is a property of *events that have not happened yet*, about which a run of zeros says nothing, so SSE gets no such revision — its entire retained state is $\Lambda(t)$. At matched expected further cases, SSE’s residual risk is a rare burst and SSI’s a more likely singleton, and the question “any further case?” sees the second as the larger risk. The contrast is therefore not DLO against the other mechanisms: it is how many units the same $\Lambda(t)$ is divided over and whether each unit’s dispersion survives to time t . DLO divides over future days, SSI over past individuals, SSE not at all.
- **The ordering is a property of the question as much as of the models.** Asking instead whether transmission would re-establish reverses SSE and SSI on these same data, because a cluster seeds a new chain far more readily than a lone case. Both orderings are correct answers to different questions, and the end-of-outbreak criterion in use asks the first.
- **The model probabilities strongly favour SSI** — 1.00 against 4×10^{-6} for DLO and 3×10^{-10} for SSE, from log evidences of -84.8 , -97.1 and -106.8 nats. At a fixed k this partly reports that 0.18 is a better value for SSI than for the other two, which is the transplantation error restated rather than independent evidence about mechanism. Analysis 2 separates the two readings.
- **R_{post} is comparable across the three models** — 0.59 (0.30–1.20), 0.68 (0.34–1.48) and 0.72 (0.37–1.50) (posterior median and 95% credible interval) — so the divergence in the risk curves does not come from a different inferred effect of the intervention. The corresponding R_{pre} estimates are 2.55 (1.20–6.05), 1.81 (0.72–5.35) and 1.63 (0.74–3.86).
- **Monte-Carlo error does not account for the separation.** Over the case-free tail in which the crossings occur, the largest Monte-Carlo standard error on any of the three curves is 0.0041, two orders of magnitude below the differences above.

6.2 Mechanism, with the dispersion estimated

- **The main split in k is between individual- and day-level heterogeneity.** The posterior medians (95% credible intervals) are 0.378 (0.230–0.629) for DLO, 0.502 (0.307–0.853) for SSE and 0.141 (0.075–0.273) for SSI. DLO against SSI: overlap 0.095, median ratio 2.68, $\Pr(k_{\text{DLO}} > k_{\text{SSI}}) = 0.990$. SSE against SSI: overlap 0.032, ratio 3.56, probability 0.999. But DLO against SSE: overlap 0.589 and $\Pr(k_{\text{DLO}} > k_{\text{SSE}}) = 0.219$ — the two are barely distinguishable. **It is SSI that is the outlier, not DLO.**
- **The split is day-level against individual-level, which is where onset-as-infection bites hardest.** DLO and SSE both attach their excess variance to a *day*: DLO to the day’s aggregate incidence, SSE to the day’s pooled transmission through a freshly drawn λ_t . SSI attaches it to the *individual*. The series being fitted is onsets, and an incubation period of mean 11.4d and standard deviation 8.1d scatters each day’s infections forward over a wide kernel — so a model reading onsets as infections sees day-to-day variation that the convolution has already largely averaged out, and answers with a larger k (less overdispersion) than the infection process carried. SSI is close to invariant to this, because the convolution *regroups* individuals across days while the aggregate infectivity of n independent individuals is $\text{Gamma}(kn, k)$ whichever n they are: regrouping moves the cohorts, not the variance. Because all three models here are infection-anchored, this analysis alone cannot separate that explanation from the structural difference of Section 4.4; the onset-anchored comparison

Équateur 2018: naive (onset-as-infection) models, k estimated

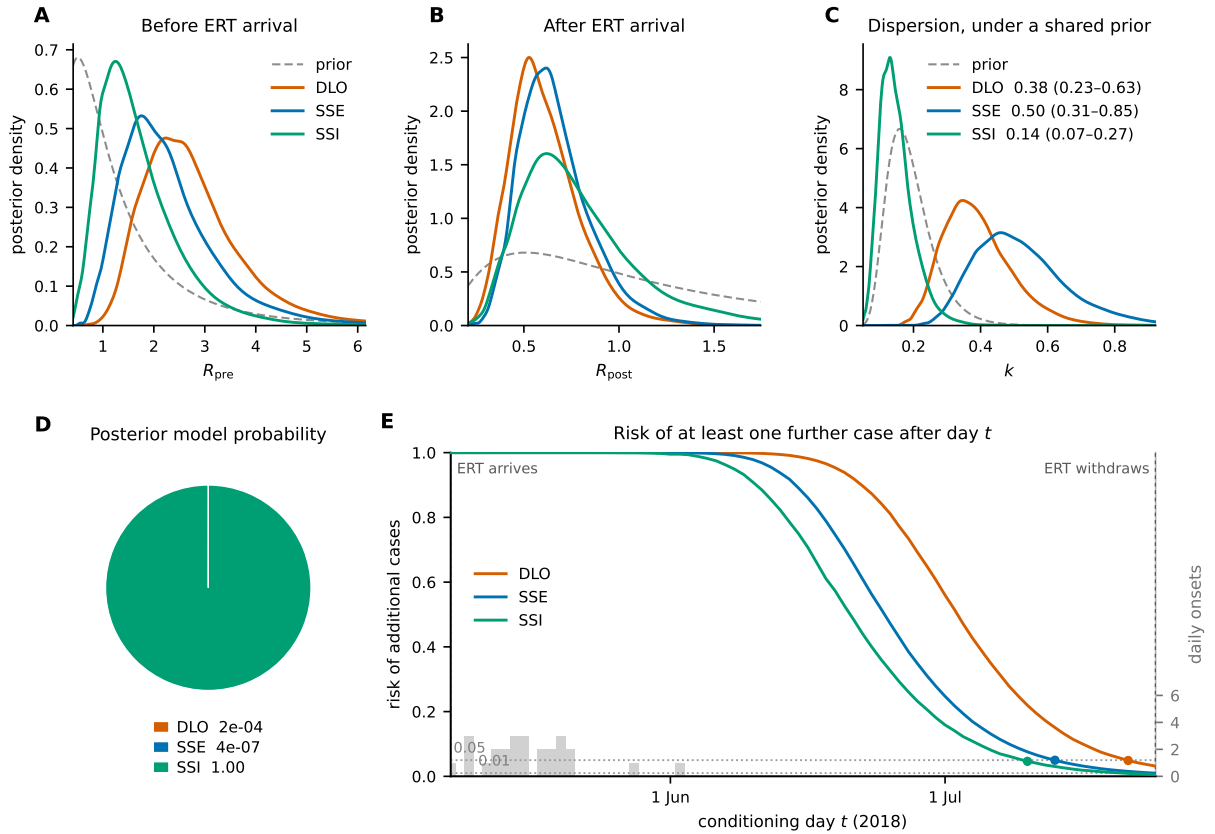


Figure 2: The models infer different dispersion parameters, and their risk curves remain separated. As Fig. 1, with k estimated. **C**: the three k posteriors over the prior they share, with each model’s posterior median and 95% credible interval in the legend. **D**: posterior model probabilities. **E**: risk of additional cases, with the view starting at the ERT’s arrival because from there until well past the last case all three curves sit at 1. The curves are computed over the whole window either way; only the view is trimmed.

provides that test.

- **Only SSI's posterior is compatible with the literature value.** SSI's 95% credible interval, (0.075, 0.273), contains 0.18; DLO's (0.230, 0.629) and SSE's (0.307, 0.853) do not, and both medians sit above the prior's own 97.5th percentile of 0.36, despite the informative prior.
- **The evidence says the same thing more sharply.** Letting k move is worth +7.41 nats to SSE and +3.90 to DLO, but only +0.21 to SSI. Analysis 1 was charging DLO and SSE for a value poorly supported under those models, whereas 0.18 is compatible with SSI on this series.
- **The risk gap persists when k is estimated.** At each model's inferred k , DLO still crosses 0.05 on day 107, against day 99 for SSE and day 96 for SSI, and still **never** reaches 0.01 inside the window, where SSE does on day 110 and SSI on day 107. Misapplying a literature k creates the gap; the day-level mechanism sustains it.
- **The posterior model probabilities keep their direction** — SSI 1.00, DLO 2×10^{-4} , SSE 4×10^{-7} . Because each model has its own inferred dispersion, this comparison is not confounded by imposing the same numerical value of k .

6.3 Onset anchoring

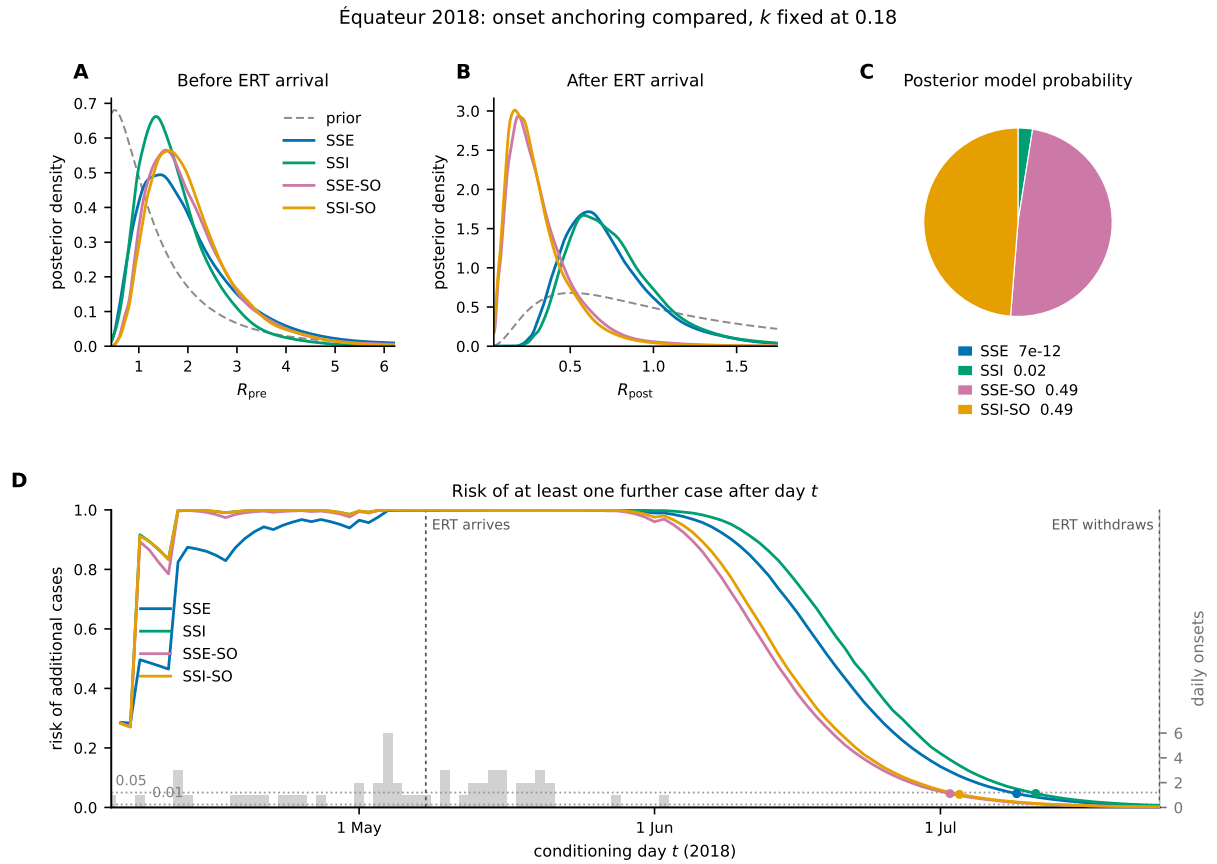


Figure 3: Onset anchoring at a fixed dispersion parameter. SSE, SSI, SSE-SO and SSI-SO with k held at 0.18; panels as in Fig. 1, with a uniform prior over 4 models in C. All four models are driven by the same onset-to-onset serial interval, so the comparison isolates anchoring.

- **Onset anchoring advances both crossings by 8–12 days.** With k estimated (Fig. 4), SSE-SO crosses 0.05 on day 88 against SSE's day 99, an advance of 11 days, and 0.01 on day 99 against day 110, an advance of 11 days. SSI-SO advances SSI's crossings by 7 and 8

Équateur 2018: onset anchoring compared, k estimated

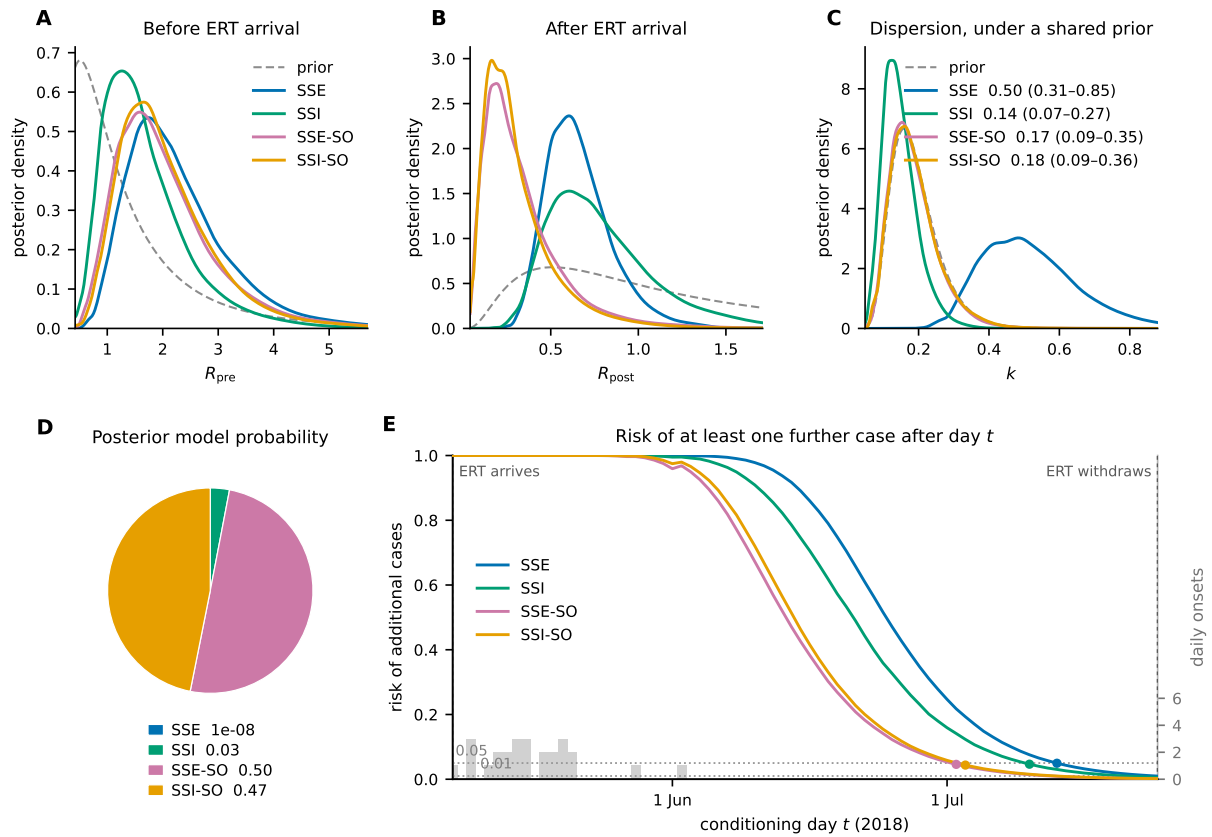


Figure 4: Onset anchoring with the dispersion estimated. Panels as in Fig. 2. In panel C, SSE-SO’s k posterior shifts below SSE’s and towards the literature estimate, whereas SSI-SO remains close to SSI. This contrast supports attenuation of day-level, but not individual-level, heterogeneity when onsets are treated as infections.

days respectively (day 89 against day 96, and day 99 against day 107). At a fixed k (Fig. 3) the same comparisons give 7 and 7 days for SSE-SO against SSE, and 8 and 8 days for SSI-SO against SSI. The displacement is on the scale of the mean incubation period, 11.4 d.

- **Why the direction is *earlier*, and why it is not a shorter generation time.** All four models share the same onset-to-onset serial interval; what changes is which side of the conditioning day a transmission falls on once that interval is cut into transmission and incubation pieces. Proposition 3 makes this explicit: the onset-anchored models place infection at transmission time, so infections already generated under the historical R_{post} enter $M(t)$ and stay in the incubation pipeline when control is reset, while the remaining TOST opportunity $W(t)$ of older onset cohorts is already largely spent. The infection-anchored models have no pipeline, move the infection decision to the later symptom-onset date, and apply the reset R^* to it, which makes the residual state look younger and drain later.
- **The infection-anchored convention more than doubles the inferred R_{post} .** With k estimated the posterior medians (95% credible intervals) are 0.63 (0.37–1.11) for SSE and 0.73 (0.36–1.63) for SSI, against 0.28 (0.08–0.96) for SSE-SO and 0.26 (0.08–0.89) for SSI-SO. The infection-anchored convention applies the day-33 switch to *onsets*, so post-arrival onsets that arose from pre-arrival infections are charged to the controlled period. R_{pre} is far more stable across the four — 2.09 (0.98–4.77), 1.52 (0.67–3.72), 1.86 (0.82–4.58) and 1.88 (0.86–4.28) — consistent with an effect on *when* the switch lands rather than on the epidemic’s early growth.
- **The results support attenuation of day-level dispersion.** SSE-SO’s k posterior, 0.174 (0.089–0.347), sits below infection-anchored SSE’s 0.501 (0.306–0.847) and towards the literature value: median ratio 2.87, overlap 0.086, $\Pr(k_{\text{SSE}} > k_{\text{SSE-SO}}) = 0.992$. SSI moves far less — 0.140 (0.072–0.267) against SSI-SO’s 0.176 (0.089–0.356), overlap 0.737. Incubation convolution attenuates *day-level* dispersion when onsets are read as infections; individual-level dispersion is comparatively stable under regrouping.
- **The two onset-anchored mechanisms infer similar k values.** SSE-SO and SSI-SO agree almost exactly — overlap 0.983, median ratio 0.99, $\Pr(k_{\text{SSE-SO}} > k_{\text{SSI-SO}}) = 0.492$ — where their infection-anchored counterparts differ by a factor of 3.58 at an overlap of 0.031. The event-versus-individual gap of Section 6.2 is therefore largely attributable to temporal conflation in this outbreak.
- **The onset-anchored models are strongly preferred, and are indistinguishable from each other.** With k estimated the posterior model probabilities are 0.50 for SSE-SO and 0.47 for SSI-SO, against 0.03 for SSI and 1×10^{-8} for SSE; the log evidences are -81.7 , -81.8 , -84.5 and -99.4 nats. The fixed- k analysis gives the same ordering (0.49, 0.49, 0.02, 7×10^{-12}).
- **Estimating k adds little evidence for the onset-anchored models.** Releasing k is worth $+0.07$ nats to SSE-SO and 0.00 to SSI-SO — within their own estimation error — against $+7.41$ for infection-anchored SSE. Their k posteriors already sit on 0.18, so fixing it there costs them nothing.
- **The 8–12-day advance is specific to this analysis.** Its direction is conditional on the reset and switch conventions, the outbreak history and the posterior; it is not a general result for arbitrary time series or interventions.

6.4 Cases, transmission and sustained transmission

Figure 5 compares all three risks at the transplanted dispersion parameter; Fig. S2 repeats the comparison with k estimated.

- **Sustained transmission changes the ranking of the two heterogeneity mechanisms.** At the reference day under fixed k , SSE has RST = 0.038 against 0.021 for SSI, even though

Table 1: Threshold crossings of the risk of additional cases across the four analyses: the first day on which the curve settles below the threshold, with the calendar date. The ERT withdrew on day 110 (24 July 2018); “never” means the curve had not settled below the threshold by then.

Analysis	Model	RAC < 0.05		RAC < 0.01	
		day	date	day	date
1: infection-anchored, $k = 0.18$	DLO	107	21 July 2018	never	never
	SSE	95	9 July 2018	106	20 July 2018
	SSI	97	11 July 2018	108	22 July 2018
2: infection-anchored, k estimated	DLO	107	21 July 2018	never	never
	SSE	99	13 July 2018	110	24 July 2018
	SSI	96	10 July 2018	107	21 July 2018
3: onset, $k = 0.18$	SSE	95	9 July 2018	106	20 July 2018
	SSI	97	11 July 2018	108	22 July 2018
	SSE-SO	88	2 July 2018	99	13 July 2018
	SSI-SO	89	3 July 2018	100	14 July 2018
4: onset, k estimated	SSE	99	13 July 2018	110	24 July 2018
	SSI	96	10 July 2018	107	21 July 2018
	SSE-SO	88	2 July 2018	99	13 July 2018
	SSI-SO	89	3 July 2018	99	13 July 2018

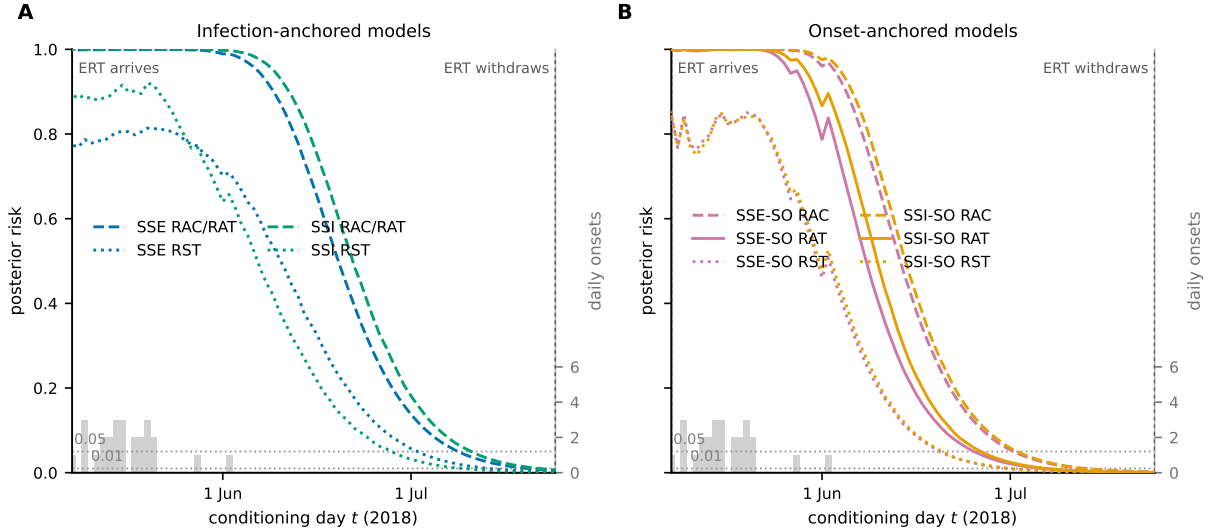


Figure 5: The risk of sustained transmission distinguishes a future transmission from a transmission lineage that persists. Models fitted with k fixed at 0.18. **A:** infection-anchored SSE and SSI; RAC = RAT by construction (dashed), while RST is dotted. **B:** onset-anchored SSE-SO and SSI-SO; dashed, solid and dotted curves are RAC, RAT and RST, respectively. Colours identify models in both panels. Grey bars show daily symptom onsets; vertical lines mark ERT arrival and withdrawal, and horizontal lines mark the 0.05 and 0.01 decision thresholds. Each point is a separate posterior fit to the record through that conditioning day, followed by a reset to R_{pre} . DLO is omitted because its day-level negative-binomial variation does not define individual transmission lineages.

their RAC values are 0.093 and 0.122, respectively. SSE assigns the remaining risk to rarer, larger bursts: a burst is less likely to occur, but more likely than an isolated transmission to start a lineage that survives. Thus the ordering measured for RAC is not an ordering of late-outbreak risk in general; it depends on the event being asked about.

- **Onset anchoring advances the decline in sustained-transmission risk.** At fixed k , SSE-SO settles below 0.05 on day 76, against day 89 for SSE; SSI-SO does so on day 77, against day 84 for SSI. The normalised delay distributions affect when residual transmission and the incubation pipeline cross the conditioning day, but not the eventual-extinction root under the constant reset reproduction number.
- **The pipeline is worth about a quarter of a unit of probability at its widest.** With k estimated, RAC exceeds RAT by at most 0.273 for SSE-SO, on day 65 (9 June 2018), and by at most 0.236 for SSI-SO, on day 67. The gap peaks shortly after the last observed onset (day 58), when the pipeline is largest relative to the transmission still to come, and closes as it drains.
- **It delays a case-based declaration by most of a week.** With k estimated, SSE-SO's RAT settles below 0.05 on day 82 and below 0.01 on day 92, against day 88 and day 99 for RAC — later by 6 and 7 days. For SSI-SO the same comparison gives 6 and 6 days (RAT on days 83 and 93 against RAC on days 89 and 99).
- **The two quantities answer different questions, and the difference is decision-relevant.** RAT isolates what relaxing interventions does; RAC is what a surveillance system would actually see. A criterion written on cases keeps interventions in place while the pipeline drains, even though transmission risk has already settled — and an infection-anchored model, which identifies the two by assumption, cannot represent that distinction at all.
- **Fixing k changes very little here,** as the near-identical panels show: the widest gap is 0.276 for SSE-SO and 0.237 for SSI-SO, against 0.273 and 0.236 when it is estimated — a consequence of the onset-anchored k posteriors already sitting on the literature value.

6.5 Incomplete reporting

Every result above reads the observed onset series as the true one. A day with no reported onset is then a day with no onset, and it is exactly those days that drive the risk down. Relaxing the assumption asks what a case-free record is worth when cases can go unseen.

The observation model is the one of Section 4.1 applied to latent true onsets D_t , with $c_t \mid D_t \sim \text{Binomial}(D_t, \pi)$ reported. The estimand is unchanged and still concerns *true* cases: $\text{RAC}(t)$ is the posterior probability of at least one further true onset, whether or not anyone would record it. Only $D_0 = c_0$ is fixed, so no day after the first is constrained to have been fully observed. π is an assumed input, not an estimate — these data cannot identify it without external information — so Fig. 6 is a sweep over assumed values rather than a fit.

- **Unreported cases can only add risk, and they do so throughout.** The 60% and 80% curves lie above the completely-reported one on every conditioning day, for both models. This is a check as much as a result: thinning cannot remove risk, so a crossing would indicate an error rather than a finding.
- **The delay is on the order of a week, and it grows as the assumption is weakened.** SSE-SO first falls below 0.05 on 2 July 2018 under complete reporting, 7 July 2018 at 80% reporting, and 11 July 2018 at 60%. For SSI-SO the same three dates are 3 July 2018, 7 July 2018 and 12 July 2018.
- **The stricter threshold is not reached at all under the weakest assumption.** At 0.01, SSE-SO settles on 13 July 2018 under complete reporting and 18 July 2018 at 80%, but at 60% it does not reach the threshold within the analysis window at all: the risk on the final day is still 0.011, against 0.002 under complete reporting. SSI-SO behaves the same way. A declaration criterion written at 0.01 is therefore not robust to a reporting

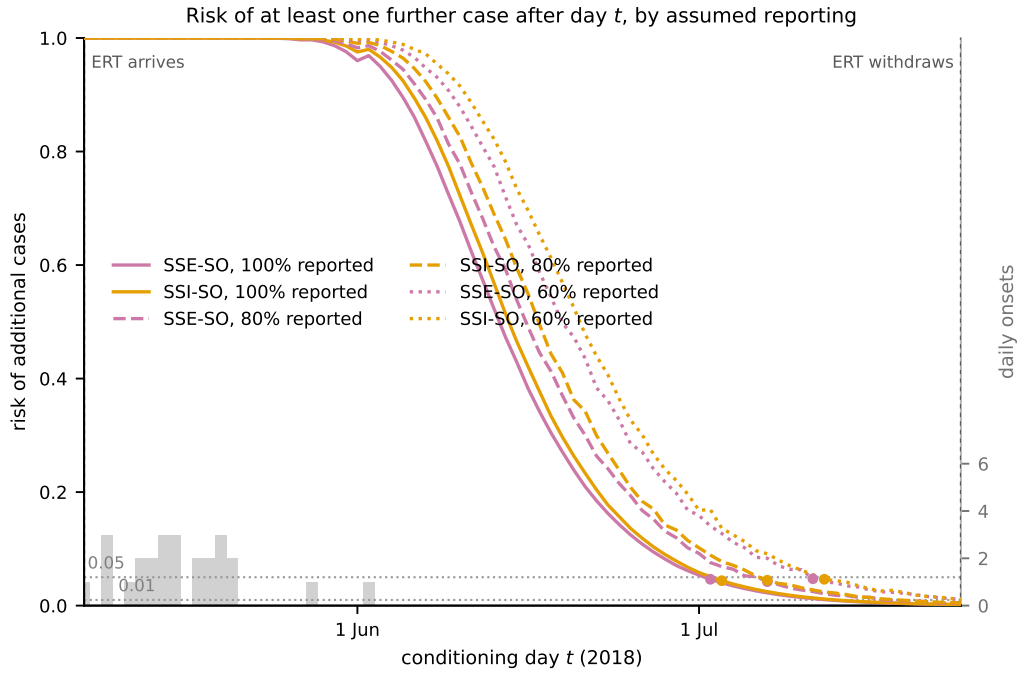


Figure 6: Assuming cases went unreported delays the declaration, and by more than the choice between mechanisms does. RAC for SSE-SO (pink) and SSI-SO (orange) under complete reporting (solid), 80% reporting (dashed) and 60% reporting (dotted), from the ERT's arrival onward. k is held at 0.18 throughout, so the only thing varying between linestyles is the assumed reporting probability. Filled circles mark where each curve first falls below 0.05 and stays there. Grey bars show daily *reported* onsets; the fitted models carry a posterior over the true counts behind them. Each point is a separate real-time fit to the reported record through that conditioning day.

assumption this weak, on a record of this length.

- **Under-reporting moves the curves further than the choice of mechanism does.** The two models' 0.05 crossings differ by at most a day at any one reporting level; dropping from complete reporting to 60% moves each of them by well over a week. Where surveillance completeness is uncertain, it dominates the model-structure question this report otherwise concerns itself with.
- **The cost is in the sampling, not the model.** With the totals random, nothing in the latent block can be marginalised — SSE-SO carries 110 sampled latents against 58 under complete reporting — and the true counts add a discrete block of their own. Convergence is still adequate across all 156 fits at 60% reporting (worst $\hat{R}$ 1.016, smallest bulk ESS 353), but the Monte-Carlo standard error on the late curve is correspondingly larger: 0.0076 against 0.0020 under complete reporting.

A reporting *delay* from onset to report is implemented alongside this and would right-truncate recent days automatically, one conditioning day at a time. It is not used here: the Équateur line list gives onset dates rather than reporting dates, so there is nothing to estimate the delay from.

7 Planned sensitivity analyses

The following analyses remain to be completed before drawing conclusions about robustness.

1. **Outbreak-specific serial interval** (mean 19.46 d, standard deviation 6.08 d) — *infection-anchored models only*. Its lighter tail may reduce the estimated risk after a long case-free period. It is incompatible with the onset-anchored models under the independent incubation/TOST decomposition because its variance is smaller than the incubation variance (Section 2.2).
2. **Incubation/TOST decomposition.** Re-estimate SSE-SO and SSI-SO under alternative published Ebola incubation distributions while holding the onset-to-onset serial interval fixed. This will quantify dependence on a decomposition that the serial interval alone does not identify.
3. **Retrospective rather than real-time conditioning.** Retain the state at day t from a fit to the complete record, as an explicitly retrospective counterpart to the estimand of Section 4.1. This asks a different question — what the evidence now says about a past decision, rather than what was knowable at the time — and the gap between the two answers is itself the quantity of interest.
4. **Shifted R -switch for the infection-anchored models.** The onset-anchoring comparison confounds model structure with the indexing convention for the intervention. Moving the switch in the infection-anchored models back by $\mathbb{E}[A] = 11.4$ days will help separate the effect of intervention indexing from the effect of model structure.
5. **Non-empty initial incubation pipeline** for the onset-anchored models. Seed one or two already-infected individuals at day 0 with residual incubation drawn from f^{inc} , and measure the effect on R_{pre} and the risk curve. This assesses sensitivity to whether the first observed case is the epidemiological index case (Section 2.1).
6. **Prior sensitivity**, particularly for the informative prior on k , whose interpretation differs between models.

A The two forms of the onset-anchored models are equivalent

This appendix proves Proposition 1 and gives the additional argument needed for (13). Throughout, the explicit infection-process form is as defined in Section 3.3: infections J_t are drawn from the model’s offspring distribution, and the J_u infections of day u are allocated to onset days by the incubation split (9), of which only the totals D_t are observed.

Proof of Proposition 1

By hypothesis $J_t \mid E_t \sim \text{Poisson}(E_t)$, with E_t measurable after the day- t onset cohort and any associated latent state have been determined.

Step 1 — Poisson thinning. Multinomial splitting of a Poisson count yields *independent* Poisson parts: conditional on E_u , $N_{u,a} \sim \text{Poisson}(E_u f_a^{\text{inc}})$ independently over $a \geq 1$, with $J_u = \sum_a N_{u,a}$ recovered as their sum. The latent J_u therefore carries no information beyond the parts and can be dropped. (This is also the step that makes the incubation pipeline of (25) Poisson.)

Step 2 — Poisson superposition. $D_t = \sum_{a \geq 1} N_{t-a,a}$ is a sum of independent Poissons, hence $D_t \mid (E_u)_{u < t} \sim \text{Poisson}(\sum_{a \geq 1} f_a^{\text{inc}} E_{t-a})$, which is (10).

Step 3 — the joint law, not just the marginal. Because f^{inc} has no mass at lag 0, E_{t-a} for $a \geq 1$ depends only on onset cohorts and associated states dated no later than $t-1$, so the display above is precisely the conditional law $p(D_t \mid D_{0:t-1})$. The parts $\{N_{u,a} : u+a=t\}$ composing D_t are disjoint from those composing any other $D_{t'}$, so distinct days’ onsets are conditionally independent given the E s. Chaining $p(D_{0:T}) = \prod_t p(D_t \mid D_{0:t-1})$ gives equality of the full joint laws. $\square$

Why the hypothesis covers all three models

Cori-SO and SSI-SO satisfy it directly, with $E_t = R_t \lambda_t$ and $E_t = R_t \lambda_t^Y$ respectively; under SSI-SO the Gamma infectivities are explicit in both forms and the thinning and superposition arguments do not touch them.

SSE-SO needs one further step, because its offspring law (13) is negative binomial rather than Poisson. If $\tilde{\lambda}_t \mid \lambda_t \sim \text{Gamma}(k\lambda_t, k)$ and $J_t \mid \tilde{\lambda}_t \sim \text{Poisson}(R_t \tilde{\lambda}_t)$, then $R_t \tilde{\lambda}_t \sim \text{Gamma}(k\lambda_t, k/R_t)$ and, marginally, $J_t \sim \text{NB}(\text{mean} = R_t \lambda_t, \text{disp} = k\lambda_t)$ — which is exactly (13). So (14) merely *un-marginalises* the negative binomial into an explicit per-day Gamma random effect, after which J_t is Poisson conditional on $\tilde{\lambda}_t$ and the proposition applies with $E_t = R_t \tilde{\lambda}_t$. The $\tilde{\lambda}_t$ are independent across t given (λ_t) , and each λ_t is a function of $D_{\leq t}$, so the day- t conditional still depends only on $D_{0:t-1}$ once $\tilde{\lambda}_{< t}$ is integrated out. $\square$

B Data and code availability

The accompanying repository contains the daily onset series, analysis configuration, source code, posterior samples, derived quantities, figures and this manuscript. The full analysis is reproducible from the onset data through a documented Snakemake workflow. Numerical values in the manuscript are generated from the saved analysis outputs so that text and figures remain synchronised. The repository also includes the simulation, particle-method and marginal-likelihood validation analyses described in Section 5.6.

Because the estimand conditions on the record through each day in question, the risk curves are the expensive step rather than a summary of a single fit, and the workflow reflects that: each curve is produced by one Markov chain Monte Carlo fit per conditioning day, and the sampler diagnostics for every one of those fits are written alongside the curve. The workflow refuses to produce a curve if any of its fits fails the convergence criteria of Section 5.2. The comparison estimator of Section 4.5, which fits the complete record once and re-derives the latent states by particle filtering, is available through the same configuration and is used only for the comparison reported there.

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Supplementary figures

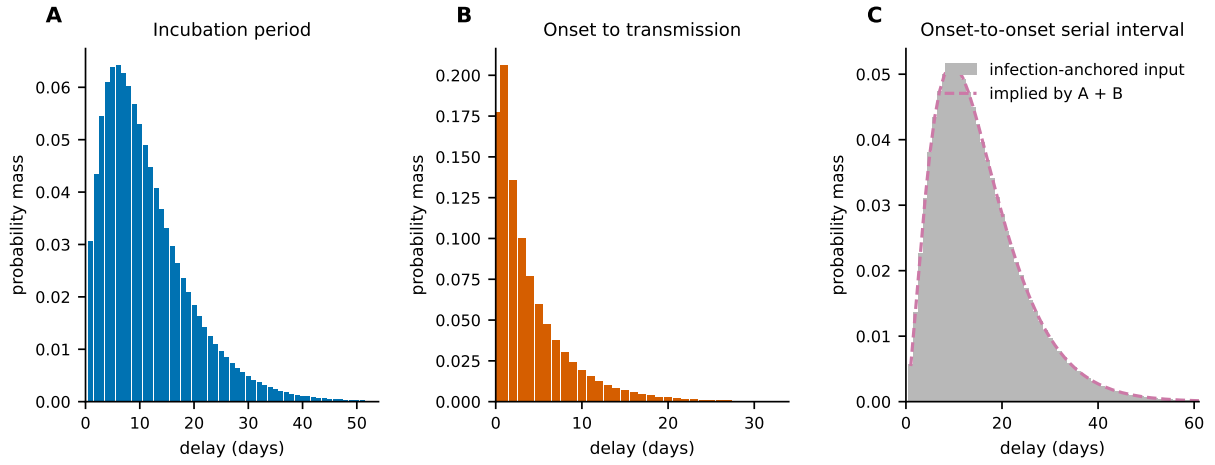


Figure S1: Delay distributions used by the infection- and onset-anchored models. **A**, incubation-period distribution from infection to symptom onset. **B**, time from the infector's symptom onset to transmission (TOST). **C**, the discretised onset-to-onset serial interval used directly by the infection-anchored models and the serial interval implied for the onset-anchored models by the discrete convolution of the distributions in **A** and **B**.

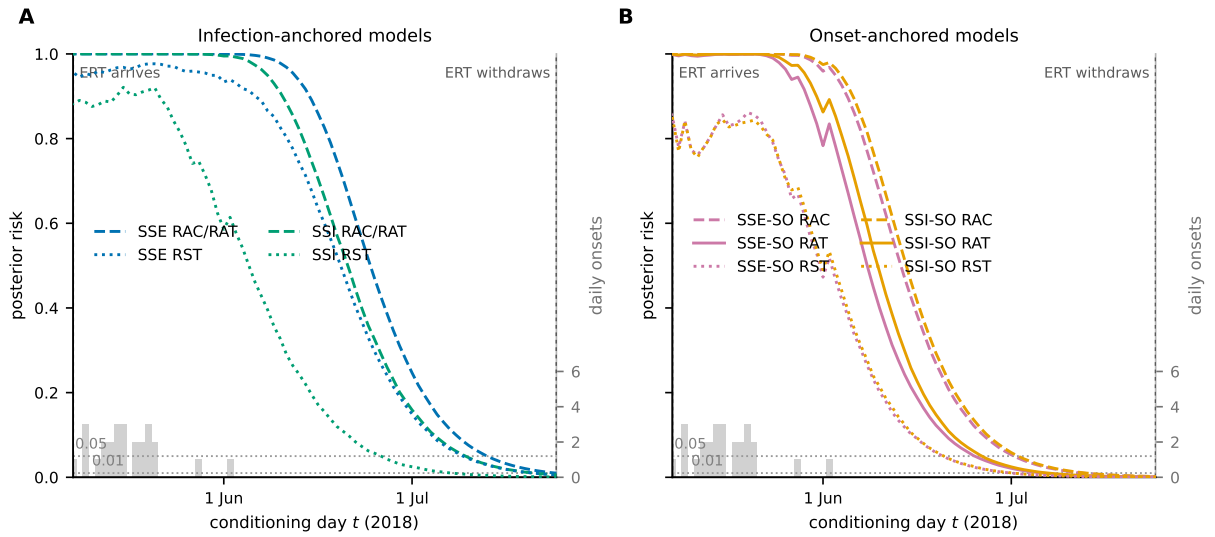


Figure S2: Risks of additional cases, additional transmission and sustained transmission with the dispersion estimated. Layout and line conventions are identical to Fig. 5. **A**: infection-anchored SSE and SSI. **B**: onset-anchored SSE-SO and SSI-SO. The dotted RST curves use each posterior draw's extinction probability under the reset to R_{pre} ; they are zero for subcritical reset draws.